

Márk Gyulai

Senior Java Developer

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About me

I develop backend systems based on Java and Kotlin, with a particular focus on sustainable architecture, maintainability and long-term evolvability. As an AI consultant, I help companies identify the areas where artificial intelligence can create real business value, and I support its practical, goal-oriented adoption.

Experience

Adroit Group Senior Software Engineer

October 2025 – present

I work on three different projects, two of which are Kotlin-based. Alongside backend development, I take part in technical design, API design, testing and maintaining code quality. My work also includes exploring practical applications of AI and implementing the related developments.

- **Car rental system** (Kotlin, Spring Boot, Spring Security, PostgreSQL, Exposed, Jenkins, GitLab, Jira, Confluence, Claude): I work on the backend of a car rental service with Kotlin and Spring Boot.
  - Implementing business logic and more complex calculations, processing and aggregating data
  - Building asynchronous synchronization with a related subsystem
  - Managing the PostgreSQL database and designing transactions properly
  - Implementing an OAuth-based authentication flow and token handling with Spring Security
  - Developing scheduled background jobs with Spring's @Scheduled support
  - Designing APIs with a documentation-first (design-first) approach
  - Writing unit and integration tests, and manual testing
  - Regular meetings with the client about development tasks and expectations
- **Financial application** (Kotlin, Spring Boot 4, OpenAPI, GitLab, ArgoCD, Jira, Confluence, Claude): As a member of an international team, I took part in developing the backend-for-frontend (BFF) application of a Swiss fintech startup, using Kotlin and Spring Boot 4.
  - Designing APIs with a documentation-first approach, based on an OpenAPI specification
  - Integrating the backend with the frontend and the related subsystems
  - Supporting contract-based development with artifacts generated from the OpenAPI specification
  - Working with an international development team on designing and implementing solutions
- **AI consulting and software development** (C#, .NET, Node.js, Angular, PostgreSQL, AI, MCP, Prompt engineering, Azure DevOps, Jira, Confluence, Claude): As an AI consultant and developer, I explore the practical applications of artificial intelligence and implement the related solutions. In this work we examine business needs, technical feasibility and the options for integration into existing systems alike.
  - Developing a platform that supports manual testing, with a C#/.NET backend, a PostgreSQL database, Node.js and Angular
  - Building an MCP server that lets the platform connect to AI agents
  - Implementing database transactions and error-handling processes
  - Applying a code-first development approach
  - Upholding code quality and Clean Code principles in the development process
  - Building project- and repository-level AI skills
  - Writing and maintaining technical documentation
  - Working with the team on shaping technical solutions and development directions

RDI Software Hungary Software Engineer

August 2024 – October 2025

I worked on the development of an international enterprise's application as a member of a team responsible for user accounts, authentication and access management. My tasks included developing and maintaining microservices, implementing API integrations, handling database operations and testing. Within the team I took part in code reviews, and I also provided professional support to a junior developer.

- **User accounts and authentication** (Java, Microservices, REST API, OpenAPI, SQL, NoSQL, Python, Authentication, Authorization, Unit testing, GitHub, ArgoCD, Jira, Confluence, GitHub Copilot): In an international team, I developed backend services for user accounts, authentication and authorization.
  - Maintaining and improving existing microservices
  - Designing and implementing new microservices
  - Designing REST API endpoints, structuring the OpenAPI specification and creating reusable schemas
  - Investigating bugs by analysing metrics, traces and logs
  - Raising test coverage above 90%
  - Restoring NoSQL data corrupted during migrations with Python scripts

- Managing database transactions
- Taking part in code reviews
- Integrating Google sign-in with the Google SDK

EPAM Systems Software Engineer
January 2022 – August 2024

As a member of an international development team, I worked on the development of a financial platform. On this project I deepened my knowledge of multithreaded programming, the Domain-Driven Design (DDD) approach and functional programming. My tasks included testing, integration with external systems, fixing concurrency bugs and managing database transactions. I also took part in improving test coverage and mentoring a junior developer.

- **Financial platform – Web framework serving widgets** (Java, Spring, Spring Boot, Spring MVC, Spring Security, MongoDB, AWS, Docker, Podman, JUnit, Mockito, Maven, Git): I worked on the development of a framework that hosts and serves the web widgets of a financial platform. I took part in developing and maintaining the framework, and in ensuring the reliable operation of the platform.
  - Investigating and fixing bugs
  - Writing new unit tests
  - Implementing new features
  - Updating the database
  - Taking part in code reviews
- **Financial platform – Backend development for web widgets** (Java, Spring, REST API, SQL, AWS, Git, Maven, Jenkins, JUnit, Mockito): I worked on the backend development of a financial platform's web widgets. My tasks included developing and maintaining the backend features serving the widgets, supporting the REST APIs and data handling, and investigating and fixing related bugs. I also took part in testing the solutions.
  - Designing and developing new features
  - Investigating and fixing bugs, including those caused by multithreading
  - Writing unit, system and acceptance tests, and modifying the acceptance test logic
  - Developing automated UI and API tests, using Selenium for the UI tests
  - Manual testing
  - Improving test coverage
  - Modifying REST endpoints and creating new ones, and creating and updating the Swagger documentation
  - Integrating other systems' APIs, with asynchronous communication where needed
  - Writing SQL queries and managing database transactions
  - Refactoring and maintaining the code base
  - Taking part in code reviews
  - Mentoring a junior developer
- **Internal wellbeing application** (Java, Spring Boot, Spring MVC, Spring Security, MongoDB, Google Cloud Platform, Docker, Podman, JUnit, Mockito, Maven, Git): I took part in improving the technical background and infrastructure of an application supporting an internal wellbeing initiative. The application supported a challenge built around physical activity. My tasks included developing the application, improving existing solutions and carrying out the related testing.

EPAM Systems Junior Software Engineer
July 2021 – January 2022

As a junior developer, I took part in the development of EPAM's internal exam system, and then continued my work on the backend development of a financial platform's web widgets. During this period I gained hands-on experience in developing, testing and maintaining enterprise Java applications.

- **Internal exam system** (Java, Spring, Hibernate, PostgreSQL, Angular, JSP, Maven, Jenkins, Git, TestNG, Mockito, VisualVM, Vagrant, Chef): I took part in developing a Spring-based internal web application used to examine participants of the Java Mentoring Program and to put the exams together. I investigated and fixed bugs, improved the setup of the virtual development environment, and improved the guides for setting up the environment and for day-to-day development work. This helped newly joining developers settle in and work independently.
  - Designing and developing new features
  - Taking part in code reviews
  - Writing unit tests
  - Manual testing
  - Improving my skill set and learning new technologies
- **Financial platform – Backend development for web widgets** (Java, Spring, REST API, SQL, AWS, Git, Maven, Jenkins, JUnit, Mockito): I took part in the backend development of a financial platform's web widgets. I developed and maintained backend features, and contributed to fixing related bugs and testing the solutions.
  - Investigating and fixing bugs
  - Modifying the acceptance test logic
  - Writing new unit, system and acceptance tests
  - Implementing new features
  - Refactoring tasks
  - Modifying REST endpoints and creating new ones
  - Creating and updating the Swagger documentation

- Writing automated UI and API tests

Skills

AI and Model Context Protocol (MCP) | Java 8–25 | Kotlin | Spring Boot | Software design and development principles | Testing | Tools | Angular

Side projects

lizacover.hu

My own portfolio

Education and certificates

Mathematics and Computer Science, BSc
University of Miskolc, 2017–2021

Java Mentored Education Program
EPAM Systems, September 2020 – July 2021

English language exam, B2

Languages
- Hungarian
- English

Driving licence
Category B

Anthropic certificates

Building with the Claude API
Anthropic, March 2026

Introduction to Model Context Protocol
Anthropic, March 2026

Introduction to agent skills
Anthropic, March 2026

Claude Code in Action
Anthropic, March 2026

More certificates (LinkedIn Learning)

Microservices Foundations · June 2021
Docker for Developers · January 2024
Introduction to Docker for Java Developers · January 2024
Java 17: First Look · January 2024
Java: Automated API Testing with REST Assured · January 2024
Learning Java 17 · October 2023
Java 8 for Professionals · October 2023
First Look: Java 10 and Java 11 · October 2020